FREE CENTRAL GENERATORS FROM THE ANTIPODE SERIES OF THE GENERAL LINEAR SUPER YANGIAN

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Editorial note. M,N>=1 and base field C are the author standing hypotheses retained from Section2. Supercenter is the definition used; the proof shows its generators are even.

Editorial note. In1305 the sum upper bound is printed M although the standalone Lie superalgebra has dimension K. Equations1286,1309,1323,1335 and1342 fix the intended full range1..K; retain and disclose the original index slip.

Editorial note. The older RTT/PBW theorem is supporting material, not a second new count. The separate quantum-Berezinian relation and its later center corollary are unselected.

Allowed standard prerequisites (editorial).

- Super linear and enveloping algebra foundations. Formal inverse series, homogeneous basis/sign rules, PBW and filtered-algebra lifting. The RTT-to-current associated graded theorem, central-series construction and the current-center lemma are retained author work, not free inputs.

Source reference. Original section 4: focused-seeds/AGP/sources/sigma-2020-112-yangian/source.tex, source lines 1374–1374. Only points to the unselected alternative quantum-Berezinian description; the current center theorem is proved entirely in the retained preceding section.

Theorem 0.1. *The elements $Z^{(2)}, Z^{(3)}, \dots$ are free generators of the centre of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$.

1. Definition of the Yangian

Throughout this article we will use the following general conventions. Let A and B be any two associative $\mathbb{Z}_2$-graded algebras. Their tensor product $\mathrm{A} \otimes \mathrm{B}$ is also an associative $\mathbb{Z}_2$-graded algebra such that for any homogeneous elements $X, X' \in \mathrm{A}$ and $Y, Y' \in \mathrm{B}$

$$(X \otimes Y)(X' \otimes Y') = XX' \otimes YY'(-1)^{\deg X' \deg Y}, \tag{1.1}$$

$$\deg(X \otimes Y) = \deg X + \deg Y. \tag{1.2}$$

For any $\mathbb{Z}_2$-graded modules U and V over A and B respectively, the vector space $U \otimes V$ is a $\mathbb{Z}_2$-graded module over $\mathrm{A} \otimes \mathrm{B}$ such that for any homogeneous elements $x \in U$ and $y \in V$

$$(X \otimes Y)(x \otimes y) = Xx \otimes Yy(-1)^{\deg x \deg Y}, \tag{1.3}$$

$$\deg(x \otimes y) = \deg x + \deg y. \tag{1.4}$$

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A homomorphism $\alpha\colon \mathrm{A} \to \mathrm{B}$ is a linear map such that $\alpha(XX') = \alpha(X)\alpha(X')$ for all $X, X' \in \mathrm{A}$. But an antihomomorphism $\beta\colon \mathrm{A} \to \mathrm{B}$ is a linear map such that for all homogeneous $X, X' \in \mathrm{A}$

$$\beta(XX') = \beta(X')\beta(X)(-1)^{\deg X \deg X'}. \tag{1.5}$$

If A is unital, let ι_h be its embedding into the tensor product $\mathrm{A}^{\otimes n}$ as the h-th tensor factor:

$$\iota_h(X) = 1^{\otimes(h-1)} \otimes X \otimes 1^{\otimes(n-h)} \qquad \text{for} \quad h = 1, \dots, n.$$

Here n can be any positive integer. We will also use various embeddings of the algebra $\mathrm{A}^{\otimes m}$ into $\mathrm{A}^{\otimes n}$ for any $m = 1, \dots, n$. For any choice of pairwise distinct indices $h_1, \dots, h_m \in \{1, \dots, n\}$ and of an element $X \in \mathrm{A}^{\otimes m}$ of the form $X = X^{(1)} \otimes \cdots \otimes X^{(m)}$ we will denote

$$X_{h_1 \dots h_m} = \iota_{h_1}\big(X^{(1)}\big) \cdots \iota_{h_m}\big(X^{(m)}\big) \in \mathrm{A}^{\otimes n}.$$

We will then extend the notation $X_{h_1 \dots h_m}$ to all elements $X \in \mathrm{A}^{\otimes m}$ by linearity.

Now let the indices i, j run through $1, \dots, M+N$. We will always write $\bar\imath = 0$ if $1 \leqslant i \leqslant M$ and $\bar\imath = 1$ if $M < i \leqslant M+N$. Consider the $\mathbb{Z}_2$-graded vector space $\mathbb{C}^{M|N}$. Let $e_i \in \mathbb{C}^{M|N}$ be an element of the standard basis. The $\mathbb{Z}_2$-grading on $\mathbb{C}^{M|N}$ is defined so that $\deg e_i = \bar\imath$. Let $E_{ij} \in \operatorname{End}\mathbb{C}^{M|N}$ be the standard matrix unit, so that $E_{ij}e_k = \delta_{jk}e_i$. The associative algebra $\operatorname{End}\mathbb{C}^{M|N}$ is $\mathbb{Z}_2$-graded so that $\deg E_{ij} = \bar\imath + \bar\jmath$.

For any n we can identify the tensor product $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes n}$ with the algebra $\operatorname{End}\big(\big(\mathbb{C}^{M|N}\big)^{\otimes n}\big)$ acting on the vector space $\big(\mathbb{C}^{M|N}\big)^{\otimes n}$ by repeatedly using the conventions (1.3) and (1.4).

Let us introduce the *Yangian* of the Lie superalgebra $\mathfrak{gl}_{M|N}$. This is the complex associative unital $\mathbb{Z}_2$-graded algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$ with the countable set of generators

$$T_{ij}^{(r)}, \qquad \text{where} \quad r = 1, 2, \dots \quad \text{and} \quad i, j = 1, \dots, M+N. \tag{1.6}$$

The $\mathbb{Z}_2$-grading on the algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$ is determined by setting $\deg T_{ij}^{(r)} = \bar\imath + \bar\jmath$ for $r \geqslant 1$. To write down defining relations for these generators we will employ the series

$$T_{ij}(u) = \delta_{ij} \cdot 1 + T_{ij}^{(1)}u^{-1} + T_{ij}^{(2)}u^{-2} + \cdots \tag{1.7}$$

in a formal variable u with coefficients from $\mathrm{Y}(\mathfrak{gl}_{M|N})$. Then for all possible indices i, j, k, l

$$(u-v)[T_{ij}(u), T_{kl}(v)](-1)^{\bar\imath\bar k + \bar\imath\bar l + \bar k\bar l} = T_{kj}(u)T_{il}(v) - T_{kj}(v)T_{il}(u), \tag{1.8}$$

where v is another formal variable. The square brackets here stand for the supercommutator. Notice that the series denoted by $T_{ij}(u)$ here and in [8] differ by the scalar factor $(-1)^{(\bar\imath+1)\bar\jmath}$.

We will also use the following matrix form of the defining relations (1.8). Take the element

$$P = \sum_{i,j=1}^{M+N} E_{ij} \otimes E_{ji}(-1)^{\bar\jmath} \in \big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes 2}. \tag{1.9}$$

This element acts on the vector space $\big(\mathbb{C}^{M|N}\big)^{\otimes 2}$ so that $e_i \otimes e_j \mapsto e_j \otimes e_i(-1)^{\bar\imath\bar\jmath}$. Here we identify the algebra $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes 2}$ with the algebra $\operatorname{End}\big(\big(\mathbb{C}^{M|N}\big)^{\otimes 2}\big)$ by using (1.3).

For any n let $\mathfrak{S}_n$ be the symmetric group acting on the set $\{1, \dots, n\}$ by permutations. For each $m = 1, \dots, n-1$ denote by σ_m the element of $\mathfrak{S}_n$ exchanging m and $m+1$. The group $\mathfrak{S}_n$ also acts on the vector space $\big(\mathbb{C}^{M|N}\big)^{\otimes n}$. This action is

defined by the assignment $\sigma_m \mapsto P_{m,m+1}$ for each m. Here we identify the algebra $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes n}$ with $\operatorname{End}\big((\mathbb{C}^{M|N})^{\otimes n}\big)$ via (1.3), (1.4).

The rational function $R(u) = 1 - Pu^{-1}$ with values in the algebra $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes 2}$ is called the *Yang R-matrix*. It satisfies the *Yang–Baxter equation* in the algebra $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes 3}(u,v,w)$

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v). \tag{1.10}$$

Since $P^2 = 1$, we also have the relation

$$R(-u)R(u) = 1 - u^{-2}. \tag{1.11}$$

Now combine all the series (1.7) into the single element

$$T(u) = \sum_{i,j=1}^{M+N} E_{ij} \otimes T_{ij}(u) \in \big(\operatorname{End}\mathbb{C}^{M|N}\big) \otimes \mathrm{Y}(\mathfrak{gl}_{M|N})\big[\big[u^{-1}\big]\big]. \tag{1.12}$$

For any n and any $p = 1, \dots, n$ we will denote

$$T_p(u) = \iota_p \otimes \mathrm{id}(T(u)) \in \big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes n} \otimes \mathrm{Y}(\mathfrak{gl}_{M|N})\big[\big[u^{-1}\big]\big]. \tag{1.13}$$

By using this notation for $n = 2$ the relations (1.8) can be rewritten as

$$(R(u-v) \otimes 1)T_1(u)T_2(v) = T_2(v)T_1(u)(R(u-v) \otimes 1). \tag{1.14}$$

Namely, after multiplying each side of (1.14) by $u - v$ it becomes a relation of series in u, v with coefficients in $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes 2} \otimes \mathrm{Y}(\mathfrak{gl}_{M|N})$ equivalent to the collection of all the relations (1.8).

Proposition 1.1. *An antiautomorphism of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *can be defined by the assignment*

$$T_{ij}(u) \mapsto T_{ij}(-u). \tag{1.15}$$

Proof. Due to the convention (1.1), by using the notation (1.13) for $n = 2$ we get

$$T_1(u)T_2(v) = \sum_{i,j,k,l=1}^{M+N} E_{ij} \otimes E_{kl} \otimes T_{ij}(u)T_{kl}(v)(-1)^{(\bar{\imath}+\bar{\jmath})(\bar{k}+\bar{l})}, \tag{1.16}$$

$$T_2(-v)T_1(-u) = \sum_{i,j,k,l=1}^{M+N} E_{ij} \otimes E_{kl} \otimes T_{kl}(-v)T_{ij}(-u). \tag{1.17}$$

By using the convention (1.5), the antihomomorphism property of (1.15) follows from the relation

$$(R(u-v) \otimes 1)T_2(-v)T_1(-u) = T_1(-u)T_2(-v)(R(u-v) \otimes 1),$$

which is obtained from (1.14) by using (1.11). The antihomomorphism (1.15) is clearly involutive and therefore bijective. □

For all indices i, j define the series $T'_{ij}(u)$ by using the element inverse to (1.12) so that

$$T(u)^{-1} = \sum_{i,j=1}^{M+N} E_{ij} \otimes T'_{ij}(u). \tag{1.18}$$

Proposition 1.2. *An antiautomorphism of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *can be defined by the assignment*

$$T_{ij}(u) \mapsto T'_{ij}(u). \tag{1.19}$$

Proof. Similarly to (1.17), by using the notation (1.13) for $n = 2$ we get

$$T_2(v)^{-1}T_1(u)^{-1} = \sum_{i,j,k,l=1}^{M+N} E_{ij} \otimes E_{kl} \otimes T'_{kl}(v)T'_{ij}(u).$$

Comparing this with (1.16) the antihomomorphism property of (1.19) follows from the relation

$$(R(u-v) \otimes 1)T_2(v)^{-1}T_1(u)^{-1} = T_1(u)^{-1}T_2(v)^{-1}(R(u-v) \otimes 1), \tag{1.20}$$

which is obtained by multiplying both sides of the defining relation (1.14) on the left and right by $T_2(v)^{-1}$ and then by $T_1(u)^{-1}$. The bijectivity of (1.19) follows from Proposition 2.2 below. □

Further, let τ be the antiautomorphism of $\operatorname{End}\mathbb{C}^{M|N}$ defined by the assignment

$$E_{ij} \mapsto E_{ji}(-1)^{\bar{\imath}(\bar{\jmath}+1)}.$$

Then by the definition (1.12) we have

$$\tau \otimes \mathrm{id}(T(u)) = \sum_{i,j=1}^{M+N} E_{ji} \otimes T_{ij}(u)(-1)^{\bar{\imath}(\bar{\jmath}+1)} = \sum_{i,j=1}^{M+N} E_{ij} \otimes T_{ji}(u)(-1)^{\bar{\jmath}(\bar{\imath}+1)}.$$

Proposition 1.3. *An antiautomorphism of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *can be defined by the assignment*

$$T_{ij}(u) \mapsto T_{ji}(u)(-1)^{\bar{\jmath}(\bar{\imath}+1)}. \tag{1.21}$$

Proof. Observe that $(\tau \otimes \tau)(P) = P$ and hence

$$(\tau \otimes \tau)(R(u-v)) = R(u-v). \tag{1.22}$$

Therefore the antihomomorphism property of (1.21) follows from the relation which is obtained by applying $\tau\otimes\tau\otimes\mathrm{id}$ to both sides of(1.14), see the proofs of Propositions 1.1 and 1.2 above. To prove the bijectivity of the antihomomorphism (1.21), observe that its square is given by

$$T_{ij}(u) \mapsto T_{ij}(u)(-1)^{\bar{\imath}+\bar{\jmath}}.$$

In particular, the square is an automorphism of the $\mathbb{Z}_2$-graded algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$. □

Put $T^{\sharp}(u) = \tau \otimes \mathrm{id}\big(T(u)^{-1}\big)$. Then by (1.18)

$$T^{\sharp}(u) = \sum_{i,j=1}^{M+N} E_{ji} \otimes T'_{ij}(u)(-1)^{\bar{\imath}(\bar{\jmath}+1)} = \sum_{i,j=1}^{M+N} E_{ij} \otimes T'_{ji}(u)(-1)^{\bar{\jmath}(\bar{\imath}+1)}. \tag{1.23}$$

Corollary 1.4. *An automorphism of the algebra* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *can be defined by the assignment*

$$T_{ij}(u) \mapsto T'_{ji}(u)(-1)^{\bar{\jmath}(\bar{\imath}+1)}. \tag{1.24}$$

Proof. The assignment (1.24) can also be obtained by first applying (1.21) to $T_{ij}(u)$ and then applying (1.19) to the result. Hence (1.24) defines an automorphism of the algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$ as a composition of two antiautomorphisms. □

By the definition (1.23) the homomorphism property of (1.24) is equivalent to the relation

$$(R(u-v) \otimes 1)T_1^{\sharp}(u)T_2^{\sharp}(v) = T_2^{\sharp}(v)T_1^{\sharp}(u)(R(u-v) \otimes 1), \tag{1.25}$$

which can also be obtained by applying $\tau \otimes \tau \otimes \mathrm{id}$ to both sides of (1.20) and then using (1.22).

Proposition 1.5. *The antiautomorphisms* (1.15), (1.19), (1.21) *of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *pairwise commute.*

Proof. The antiautomorphism (1.15) clearly commutes with either of (1.19), (1.21). To prove the commutativity of the latter two, consider the tensor product of the antiautomorphisms τ^{-1} and (1.21) of $\mathrm{End}\,\mathbb{C}^{M|N}$ and $\mathrm{Y}(\mathfrak{gl}_{M|N})$ respectively. This is is an antiautomorphism of the algebra $\mathrm{End}\,\mathbb{C}^{M|N}\otimes\mathrm{Y}(\mathfrak{gl}_{M|N})$, and the series (1.12) is invariant under this antiautomorphism. Hence the series (1.18) is also invariant under it. The latter invariance implies that (1.21) maps

$$T'_{ij}(u)\mapsto T'_{ji}(u)(-1)^{\bar{\jmath}(\bar{\imath}+1)}.$$

Hence (1.24) can also be obtained by first applying (1.19) to $T_{ij}(u)$ and then applying (1.21) to the result. Comparing this with the proof of Corollary 1.4 completes the argument. □

Consider the universal enveloping algebra $\mathrm{U}(\mathfrak{gl}_{M|N})$ of the Lie superalgebra $\mathfrak{gl}_{M|N}$. To avoid confusion, the element of $\mathfrak{gl}_{M|N}$ corresponding to $E_{ij}\in\mathrm{End}\,\mathbb{C}^{M|N}$ will be denoted by e_{ij}. By definition, the bracket on $\mathfrak{gl}_{M|N}$ is the supercommutator. Hence in $\mathrm{U}(\mathfrak{gl}_{M|N})$

$$[e_{ij},e_{kl}]=\delta_{jk}e_{il}-\delta_{li}e_{kj}(-1)^{(\bar{\imath}+\bar{\jmath})(\bar{k}+\bar{l})}. \tag{1.26}$$

By using the defining relations (1.8) one can demonstrate that there is a homomorphism

$$\mathrm{Y}(\mathfrak{gl}_{M|N})\to\mathrm{U}(\mathfrak{gl}_{M|N})\colon\ T_{ij}(u)\mapsto\delta_{ij}-e_{ji}u^{-1}(-1)^{\bar{\jmath}}. \tag{1.27}$$

This homomorphism is surjective. The relations (1.8) also imply that there is a homomorphism

$$\mathrm{U}(\mathfrak{gl}_{M|N})\to\mathrm{Y}(\mathfrak{gl}_{M|N})\colon\ e_{ji}\mapsto-T^{(1)}_{ij}(-1)^{\bar{\jmath}}. \tag{1.28}$$

The composition of homomorphisms (1.28) and (1.27) is the identity map $\mathrm{U}(\mathfrak{gl}_{M|N})\to\mathrm{U}(\mathfrak{gl}_{M|N})$. So (1.28) is an embedding of $\mathbb{Z}_2$-graded associative unital algebras. The homomorphism (1.27) is identical on the subalgebra $\mathrm{U}(\mathfrak{gl}_{M|N})$. It is called the *evaluation homomorphism* for $\mathrm{Y}(\mathfrak{gl}_{M|N})$.

There is a natural Hopf algebra structure on $\mathrm{Y}(\mathfrak{gl}_{M|N})$. A coassociative comultiplication homomorphism $\Delta\colon\mathrm{Y}(\mathfrak{gl}_{M|N})\to\mathrm{Y}(\mathfrak{gl}_{M|N})\otimes\mathrm{Y}(\mathfrak{gl}_{M|N})$ can be defined by the assignment

$$T_{ij}(u)\mapsto\sum_{k=1}^{M+N}T_{ik}(u)\otimes T_{kj}(u)(-1)^{(\bar{\imath}+\bar{k})(\bar{\jmath}+\bar{k})}, \tag{1.29}$$

where the tensor product is taken over the subalgebra $\mathbb{C}\big[\big[u^{-1}\big]\big]$ in $\mathrm{Y}(\mathfrak{gl}_{M|N})\big[\big[u^{-1}\big]\big]$. The counit homomorphism $\varepsilon\colon\mathrm{Y}(\mathfrak{gl}_{M|N})\to\mathbb{C}$ is defined by the assignment $T_{ij}(u)\mapsto\delta_{ij}$. The antipodal mapping $\mathrm{S}\colon\mathrm{Y}(\mathfrak{gl}_{M|N})\to\mathrm{Y}(\mathfrak{gl}_{M|N})$ is the antiautomorphism (1.19). Justification of all these definitions is similar to that in the case $N=0$ considered for instance in [6, Section 1]. Here we omit the details. Note that (1.28) is an embedding of Hopf algebras as by the above definitions

$$\Delta\colon\ T^{(1)}_{ij}\mapsto T^{(1)}_{ij}\otimes 1+1\otimes T^{(1)}_{ij},\qquad\varepsilon\colon\ T^{(1)}_{ij}\mapsto 0,\qquad\mathrm{S}\colon\ T^{(1)}_{ij}\mapsto -T^{(1)}_{ij}.$$

There are two natural ascending filtrations on the associative algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$. The first one is defined by assigning the degree r to the generator (1.6). Let $\mathrm{gr}\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ be the corresponding graded algebra.

Let us denote by $X^{(r)}_{ij}$ the image of $T^{(r)}_{ij}$ in the degree r component of $\mathrm{gr}\,\mathrm{Y}(\mathfrak{gl}_{M|N})$. Observe that the $\mathbb{Z}_2$-grading on the algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$ descends to $\mathrm{gr}\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ so

that the degree of the image is again $\bar\imath+\bar\jmath$. It follows from the relations (1.8) that these images supercommute. We shall prove that $\mathrm{gr}\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ is a free supercommutative algebra generated by these images.

Now introduce another filtration on the associative algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$ by assigning the degree $r-1$ to the generator (1.6). Let $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ be the corresponding graded algebra. Consider the latter algebra.

Let us denote by $Y_{ij}^{(r)}$ the image of $T_{ij}^{(r)}$ in the degree $r-1$ component of $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$. The $\mathbb{Z}_2$-grading on the algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$ descends to $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ so that the degree of the image is $\bar\imath+\bar\jmath$. The graded algebra $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ inherits from $\mathrm{Y}(\mathfrak{gl}_{M|N})$ the Hopf algebra structure too. Namely, by the above definitions in the graded Hopf algebra $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ for $r\geqslant 1$

$$\Delta: Y_{ij}^{(r)}\mapsto Y_{ij}^{(r)}\otimes 1+1\otimes Y_{ij}^{(r)},\qquad \varepsilon: Y_{ij}^{(r)}\mapsto 0,\qquad \mathrm{S}: Y_{ij}^{(r)}\mapsto -Y_{ij}^{(r)}. \tag{1.30}$$

Let us now consider the polynomial current Lie superalgebra $\mathfrak{gl}_{M|N}[u]$. The elements $e_{ji}u^r$ with $r=0,1,2,\dots$ and $i,j=1,\dots,M+N$ make a basis of $\mathfrak{gl}_{M|N}[u]$. The $\mathbb{Z}_2$-grading on $\mathfrak{gl}_{M|N}[u]$ is defined by $\deg e_{ji}u^r=\bar\imath+\bar\jmath$. Take the universal enveloping algebra $\mathrm{U}(\mathfrak{gl}_{M|N}[u])$.

Proposition 1.6. *One can define a surjective homomorphism* $\mathrm{U}(\mathfrak{gl}_{M|N}[u])\to \mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ *of* $\mathbb{Z}_2$*-graded associative algebras by mapping for* $r\geqslant 0$

$$e_{ji}u^r\mapsto -Y_{ij}^{(r+1)}(-1)^{\bar\jmath}. \tag{1.31}$$

Proof. Due to (1.26) the supercommutation relations with $r,s\geqslant 0$

$$[e_{ji}u^r,e_{lk}u^s]=\delta_{il}e_{jk}u^{r+s}-\delta_{kj}e_{li}u^{r+s}(-1)^{(\bar\imath+\bar\jmath)(\bar k+\bar l)} \tag{1.32}$$

define the Lie superalgebra $\mathfrak{gl}_{M|N}[u]$. Multiplying the relation (1.32) by $(-1)^{\bar\imath\bar k+\bar\imath\bar l+\bar k\bar l+\bar\jmath+\bar l}$ and replacing the basis elements of $\mathfrak{gl}_{M|N}[u]$ there by their images under the mapping (1.31) we get

$$\big[Y_{ij}^{(r+1)},Y_{kl}^{(s+1)}\big](-1)^{\bar\imath\bar k+\bar\imath\bar l+\bar k\bar l}=\delta_{kj}Y_{il}^{(r+s+1)}-\delta_{il}Y_{kj}^{(r+s+1)}.$$

The latter relation in $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ follows from (1.8), see for instance [6, Section 1]. This proves the homomorphism property of the assignment (1.31). This homomorphism is clearly surjective and preserves the $\mathbb{Z}_2$-grading. □

It immediately follows from (1.30) that (1.31) is a homomorphism of Hopf algebras. Here we use the standard Hopf algebra structure on $\mathrm{U}(\mathfrak{gl}_{M|N}[u])$ as the universal enveloping algebra of a Lie superalgebra. We shall demonstrate that the homomorphism (1.31) is also injective.

By comparing (1.10) with (1.14) we obtain that for any $z\in\mathbb{C}$ the assignment

$$\mathrm{End}\,\mathbb{C}^{M|N}\otimes\mathrm{Y}(\mathfrak{gl}_{M|N})\big[\big[u^{-1}\big]\big]\to\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes 2}\big[\big[u^{-1}\big]\big]:\ T(u)\mapsto R(u-z) \tag{1.33}$$

defines a representation $\mathrm{Y}(\mathfrak{gl}_{M|N})\to\mathrm{End}\,\mathbb{C}^{M|N}$. More explicitly, under (1.33) for any $r\geqslant 0$

$$T_{ij}^{(r+1)}\mapsto -E_{ji}z^r(-1)^{\bar\jmath}. \tag{1.34}$$

Note that this representation of $\mathrm{Y}(\mathfrak{gl}_{M|N})$ can be also obtained from the standard representation $\mathrm{U}(\mathfrak{gl}_{M|N})\to\mathrm{End}\,\mathbb{C}^{M|N}$ by pulling back through the evaluation homomorphism (1.27) and then back through the automorphism of $\mathrm{Y}(\mathfrak{gl}_{M|N})$ defined by mapping $T_{ij}(u)\mapsto T_{ij}(u-z)$.

The comultiplication on $\mathrm{Y}(\mathfrak{gl}_{M|N})$ now allows us to define for $n=1,2,\dots$ a representation $\mathrm{Y}(\mathfrak{gl}_{M|N})\to(\mathrm{End}\,\mathbb{C}^{M|N})^{\otimes n}$ depending on $z_1,\dots,z_n\in\mathbb{C}$. This is

the tensor product of the representations (1.33) where $z = z_1, \dots, z_n$. Due to (1.12) and (1.29) under this representation

$$T(u) \mapsto R_{12}(u - z_1) \cdots R_{1,n+1}(u - z_n). \tag{1.35}$$

Proposition 1.7. *Let the complex parameters $z_1, \dots, z_n$ and the positive integer n vary. Then the kernels of the representations* (1.35) *of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *have the zero intersection.*

Proof. Take any finite linear combination of the products $T_{i_1 j_1}^{(r_1+1)} \cdots T_{i_m j_m}^{(r_m+1)} \in \mathrm{Y}(\mathfrak{gl}_{M|N})$ with $A_{i_1 j_1 \dots i_m j_m}^{r_1 \dots r_m} \in \mathbb{C}$ being the respective coefficients. In this linear combination the number $m \geqslant 0$ and indices $r_1, \dots, r_m \geqslant 0$ may vary. Consider the image of this linear combination under the representation $\mathrm{Y}(\mathfrak{gl}_{M|N}) \to \big(\operatorname{End} \mathbb{C}^{M|N}\big)^{\otimes n}$ defined by (1.35). This image depends on $z_1, \dots, z_n$ polynomially. Let A be the sum of those terms of this polynomial which have the maximal total degree in $z_1, \dots, z_n$. Let d be this degree.

Consider the second of our two ascending filtrations on the associative algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$, the corresponding graded algebra being $\operatorname{gr}' \mathrm{Y}(\mathfrak{gl}_{M|N})$. Equip the tensor product $\mathrm{Y}(\mathfrak{gl}_{M|N})^{\otimes n}$ with the ascending filtration where the degree is the sum of the degrees on the tensor factors. Then by the definition (1.29) under the comultiplication $\mathrm{Y}(\mathfrak{gl}_{M|N}) \to \mathrm{Y}(\mathfrak{gl}_{M|N})^{\otimes n}$ for $r \geqslant 0$

$$T_{ij}^{(r+1)} \mapsto \sum_{h=1}^{n} 1^{\otimes(h-1)} \otimes T_{ij}^{(r+1)} \otimes 1^{\otimes(n-h)} + \text{lower degree terms.}$$

Therefore the sum $A \in \big(\operatorname{End} \mathbb{C}^{M|N}\big)^{\otimes n}$ coincides with the image of the sum

$$\sum_{r_1 + \dots + r_m = d} A_{i_1 j_1 \dots i_m j_m}^{r_1 \dots r_m} e_{j_1 i_1} u^{r_1} \cdots e_{j_m i_m} u^{r_m} (-1)^{m + \bar{\jmath}_1 + \dots + \bar{\jmath}_m} \in \mathrm{U}(\mathfrak{gl}_{M|N}[u])$$

under the tensor product of the evaluation representations

$$\mathrm{U}(\mathfrak{gl}_{M|N}[u]) \to \operatorname{End} \mathbb{C}^{M|N} : \ e_{ji} u^r \mapsto E_{ji} z^r \tag{1.36}$$

at the points $z = z_1, \dots, z_n$. Here we used the explicit description (1.34) of the representation $\mathrm{Y}(\mathfrak{gl}_{M|N}) \to \operatorname{End} \mathbb{C}^{M|N}$ corresponding to $z \in \mathbb{C}$. Due to Proposition 1.6 it now suffices to show that when the complex parameters $z_1, \dots, z_n$ and the positive integer n vary, the kernels of the tensor products of the evaluation representations of the algebra $\mathrm{U}(\mathfrak{gl}_{M|N}[u])$ at $z = z_1, \dots, z_n$ have the zero intersection. This will also imply that the homomorphism (1.31) is injective.

Choose any $\mathbb{Z}_2$-homogeneous basis $f_1, \dots, f_{(M+N)^2}$ of $\mathfrak{gl}_{M|N}$ such that the first vector f_1 is

$$\sum_{i=1}^{M+N} e_{ii}. \tag{1.37}$$

The corresponding elements of $\mathrm{End}\,\mathbb{C}^{M|N}$ will be denoted by $F_1,\dots,F_{(M+N)^2}$. Hence $F_1=1$. The elements f_pu^r with $r=0,1,2,\dots$ and $p=1,\dots,(M+N)^2$ constitute a basis of $\mathfrak{gl}_{M|N}[u]$. Choose any total ordering of this basis which ends with the infinite sequence $\dots,f_1u^2,f_1u,f_1$. Take any finite linear combination L of the products

$$f_{p_1}u^{r_1}\cdots f_{p_m}u^{r_m}\in \mathrm{U}(\mathfrak{gl}_{M|N}[u]) \tag{1.38}$$

with $L^{r_1\dots r_m}_{p_1\dots p_m}\in\mathbb{C}$ being the coefficients. We assume that the factors in the products are arranged according to our ordering of the basis of $\mathfrak{gl}_{M|N}[u]$. Due to the supercommutation relations in $\mathrm{U}(\mathfrak{gl}_{M|N}[u])$ we assume it without any loss of generality. The basis elements of $\mathbb{Z}_2$-degree 0 may occur in any product (1.38) with a multiplicity but those of $\mathbb{Z}_2$-degree 1 may occur at most once.

Let us denote by ρ_z the evaluation representation (1.36). More generally, denote by $\rho_{z_1\dots z_n}$ the tensor product $\rho_{z_1}\otimes\cdots\otimes\rho_{z_n}$ pulled back through n-fold comultiplication on $\mathrm{U}(\mathfrak{gl}_{M|N}[u])$. Hence $\rho_{z_1\dots z_n}$ is a homomorphism of associative algebras

$$\mathrm{U}(\mathfrak{gl}_{M|N}[u])\to\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes n}:\ f_pu^r\mapsto F_pz_1^r\otimes 1^{\otimes(n-1)}+\cdots+1^{\otimes(n-1)}\otimes F_pz_n^r.$$

Suppose that $\rho_{z_1\dots z_n}(L)=0$ for every n and all $z_1,\dots,z_n\in\mathbb{C}$. We need to prove that $L=0$.

For each product (1.38) there is a number a such that $p_1,\dots,p_a>1$ but $p_{a+1},\dots,p_m=1$. This is due to our ordering of the basis of $\mathfrak{gl}_{M|N}[u]$. The numbers a for different products (1.38) may differ, and we do not exclude the case $a=0$. Let h be the maximum of the numbers a.

Suppose $n\geqslant h$. Let ω_h be the supersymmetrisation map of the tensor product $\mathfrak{gl}_{M|N}[u]^{\otimes h}$ normalised so that $\omega_h^2=h!\omega_h$. Let W be the subspace of $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes n}$ spanned by the vectors $F_{q_1}\otimes\cdots\otimes F_{q_n}$ where at least one of the indices $q_1,\dots,q_h$ is 1. If $h=0$ then this subspace is assumed to be zero. Applying the homomorphism $\rho_{z_1\dots z_n}$ to a product (1.38) with $a=h$ gives

$$(\rho_{z_1}\otimes\cdots\otimes\rho_{z_h})\big(\omega_h\big(f_{p_1}u^{r_1}\otimes\cdots\otimes f_{p_h}u^{r_h}\big)\big)\otimes 1^{\otimes(n-h)}\prod_{b=h+1}^{m}\big(z_1^{r_b}+\cdots+z_n^{r_b}\big) \tag{1.39}$$

modulo the subspace W. Applying $\rho_{z_1\dots z_n}$ to a product (1.38) with $a<h$ gives an element of W. But a linear combination of the expressions (1.39) belongs to W only if this combination is zero.

For each product (1.38) with $a=h$ there is a number $c\geqslant h$ such that $r_{h+1}\geqslant\cdots\geqslant r_c>0$ but $r_{c+1},\dots,r_m=0$. This is due to our ordering of the basis of $\mathfrak{gl}_{M|N}[u]$. Then (1.39) equals

$$(\rho_{z_1}\otimes\ldots\otimes\rho_{z_h})\big(\omega_h\big(f_{p_1}u^{r_1}\otimes\ldots\otimes f_{p_h}u^{r_h}\big)\big)\otimes 1^{\otimes(n-h)}\prod_{b=h+1}^{c}\big(z_1^{r_b}+\cdots+z_n^{r_b}\big) \tag{1.40}$$

multiplied by n^{m-c}. Let g be the maximum of the numbers c for all products (1.38) with $a=h$.

Suppose $n\geqslant g$. Consider the pairs of sequences of indices $p_1,\dots,p_m$ and $r_1,\dots,r_m$ showing in L in the products (1.38) with $a=h$. For every such a pair there is $c\in\{h,\dots,g\}$. Then we have $p_{h+1},\dots,p_m=1$ and $r_{c+1},\dots,r_m=0$. Take all different pairs of sequences $p_1,\dots,p_h$ and $r_1,\dots,r_c$ arising in this way. The expressions (1.40) corresponding to the latter pairs are linearly independent as polynomials in $z_1,\dots,z_n$ with values in $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes n}$. This is again due to our ordering of the basis of $\mathfrak{gl}_{M|N}[u]$. Here we also employ the observation that in (1.40) the image of $\rho_{z_1}\otimes\cdots\otimes\rho_{z_h}$ does not depend on the parameters $z_{h+1},\dots,z_n$ while

the product over $b = h+1, \dots, c$ in (1.40) does depend on these parameters whenever $c > h$. Therefore if $\rho_{z_1 \dots z_n}(L) = 0$ for a certain $n \geqslant g$ and for all $z_1, \dots, z_n \in \mathbb{C}$ then for each of the latter pairs

$$\sum_{m=c}^{\infty} L^{r_1 \dots r_c 0 \dots 0}_{p_1 \dots p_h 1 \dots 1} n^{m-c} = 0.$$

There are exactly m lower indices and also m upper indices in the coefficient $L^{r_1 \dots r_c 0 \dots 0}_{p_1 \dots p_h 1 \dots 1}$ above. By letting the number n vary we now prove that all these coefficients vanish. □

In the course of the proof of Proposition 1.7 we established that the homomorphism (1.31) is injective. Together with Proposition 1.6 and the observation made just after its proof this yields

Theorem 1.8. *The Hopf algebras* $\mathrm{U}(\mathfrak{gl}_{M|N}[u])$ *and* $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ *are isomorphic via* (1.31).

2. Centre of the Yangian

Here we will give a description of the centre of the algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$. An element of $\mathrm{Y}(\mathfrak{gl}_{M|N})$ is called *central* if it supercommutes with each element of $\mathrm{Y}(\mathfrak{gl}_{M|N})$. However, we will see that the central elements of $\mathrm{Y}(\mathfrak{gl}_{M|N})$ have the $\mathbb{Z}_2$-degree 0. Hence they commute with each element of $\mathrm{Y}(\mathfrak{gl}_{M|N})$ in the usual sense. Our description comes from computing the square of the antipodal mapping S of $\mathrm{Y}(\mathfrak{gl}_{M|N})$. Another description of the centre of $\mathrm{Y}(\mathfrak{gl}_{M|N})$ will be given in the next section. In that section we will also establish a correspondence between the two descriptions.

Proposition 2.1. *There is a formal power series* $Z(u)$ *in* u^{-1} *with coefficients in the centre of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *and with leading term* 1 *such that for all indices* i *and* j

$$\sum_{k=1}^{M+N} T_{kj}(u+M-N) T'_{ik}(u) = \delta_{ij} Z(u), \tag{2.1}$$

$$\sum_{k=1}^{M+N} T'_{kj}(u) T_{ik}(u+M-N) = \delta_{ij} Z(u). \tag{2.2}$$

Proof. By using the definition (1.9) introduce the element of the algebra $\big(\operatorname{End} \mathbb{C}^{M|N}\big)^{\otimes 2}$

$$Q = (\mathrm{id} \otimes \tau)(P) = \sum_{i,j=1}^{M+N} E_{ij} \otimes E_{ij}(-1)^{\overline{ij}}. \tag{2.3}$$

The image of the action of Q on $\big(\mathbb{C}^{M|N}\big)^{\otimes 2}$ is one dimensional and is spanned by the vector

$$\sum_{i=1}^{M+N} e_i \otimes e_i.$$

Here we regard Q as an element of the algebra $\operatorname{End}\big(\big(\mathbb{C}^{M|N}\big)^{\otimes 2}\big)$ by identifying this algebra with $\big(\operatorname{End} \mathbb{C}^{M|N}\big)^{\otimes 2}$ via (1.3). We also have $Q^2 = (M-N)Q$. By using the latter relation we get

$$((\mathrm{id} \otimes \tau)(R(u)))^{-1} = \big(1 - Qu^{-1}\big)^{-1} = 1 + Q(u - M + N)^{-1}. \tag{2.4}$$

The rational function of the variable u given by the equalities (2.4) will be denoted by $R^{\sharp}(u)$.

Now multiply both sides of the relation (1.14) by $T_2^{-1}(v)$ on the left and right, and then apply τ relative to the second tensor factor $\mathrm{End}\,\mathbb{C}^{M|N}$ in $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes 2}\otimes \mathrm{Y}(\mathfrak{gl}_{M|N})$. Multiplying the resulting relation on the left and right by $R^{\sharp}(u-v)\otimes 1$ yields

$$\big(R^{\sharp}(u-v)\otimes 1\big)T_1(u)T_2^{\sharp}(v) = T_2^{\sharp}(v)T_1(u)(R^{\sharp}(u-v)\otimes 1). \tag{2.5}$$

Multiplying the latter relation by $u-v-M+N$ and then setting $u=v+M-N$ we get

$$(Q\otimes 1)T_1(v+M-N)T_2^{\sharp}(v) = T_2^{\sharp}(v)T_1(v+M-N)(Q\otimes 1), \tag{2.6}$$

see (2.4). As the image of Q in $\big(\mathbb{C}^{M|N}\big)^{\otimes 2}$ is one dimensional, either side of the relation (2.6) must be equal to $Q\otimes Z(v)$ where $Z(v)$ is a certain power series in v^{-1} with coefficients from $\mathrm{Y}(\mathfrak{gl}_{M|N})$. The equality of the left-hand side and of the right-hand side of (2.6) to $Q\otimes Z(v)$ is equivalent respectively to (2.1) and (2.2). We just need to replace the variable v in (2.6) by u.

It is immediate from (1.7) and (2.1) that every coefficient of the series $Z(u)$ has $\mathbb{Z}_2$-degree 0 and that its leading term is 1. Let us prove that all these coefficients are central in $\mathrm{Y}(\mathfrak{gl}_{M|N})$.

To this end we will work with the elements (1.13) where $n=3$. By using (1.14) and (2.5),

$$\begin{aligned}
&\big(R_{13}^{\sharp}(u-v)R_{12}(u-v-M+N)\otimes 1\big)T_1(u)T_2(v+M-N)T_3^{\sharp}(v)\\
&= \big(R_{13}^{\sharp}(u-v)\otimes 1\big)T_2(v+M-N)T_1(u)T_3^{\sharp}(v)(R_{12}(u-v-M+N)\otimes 1)\\
&= T_2(v+M-N)T_3^{\sharp}(v)T_1(u)\big(R_{13}^{\sharp}(u-v)R_{12}(u-v-M+N)\otimes 1\big).
\end{aligned} \tag{2.7}$$

On the other hand, due to (1.11) we have the identity in the algebra $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes 3}(u)$

$$R_{13}(-u)P_{23}R_{12}(u) = \big(1-u^{-2}\big)P_{23}.$$

By applying to it the antiautomorphism τ relative to the third tensor factor of $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes 3}$ and then using the definition (2.4) we get

$$Q_{23}R_{13}^{\sharp}(u+M-N)R_{12}(u) = \big(1-u^{-2}\big)Q_{23}. \tag{2.8}$$

Therefore multiplying the first and third lines of the display (2.7) by $Q_{23}\otimes 1$ on the left yields

$$\begin{aligned}
&\big(1-(u-v-M+N)^{-2}\big)T_1(u)(Q_{23}\otimes Z(v))\\
&= (Q_{23}\otimes Z(v))T_1(u)\big(1-(u-v-M+N)^{-2}\big).
\end{aligned}$$

We just need to replace the variable u in (2.8) by $u-v-M+N$ and use the relation (2.6). The last relation implies that any generator $T_{ij}^{(r)}$ commutes with every coefficient of $Z(v)$. □

The square S^2 of antipodal mapping is an automorphism of the associative algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$.

Proposition 2.2. *The automorphism* S^2 *of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *is given by the assignment*

$$T_{ij}(u)\mapsto Z(u)^{-1}T_{ij}(u+M-N).$$

Proof. By the definition (1.18) for any indices i, j we have the identity

$$\sum_{k=1}^{M+N} T_{ik}(u)T'_{kj}(u)(-1)^{(\bar\imath+\bar k)(\bar\jmath+\bar k)} = \delta_{ij}. \tag{2.9}$$

Here we use the convention (1.1). By using the definition of S this identity can be written as

$$\sum_{k=1}^{M+N} T_{ik}(u)\,\mathrm{S}(T_{kj}(u))(-1)^{(\bar{\imath}+\bar{k})(\bar{\jmath}+\bar{k})} = \delta_{ij}.$$

Let us apply the antiautomorphism S to both sides of the latter identity. We get

$$\sum_{k=1}^{M+N} \mathrm{S}^2(T_{kj}(u))\,\mathrm{S}(T_{ik}(u)) = \delta_{ij} \qquad \text{or} \qquad \sum_{k=1}^{M+N} \mathrm{S}^2(T_{kj}(u))T'_{ik}(u) = \delta_{ij}.$$

By comparing the last displayed identity with (2.1) we see that for any indices k, j

$$Z(u)\,\mathrm{S}^2(T_{kj}(u)) = T_{kj}(u+M-N). \qquad \square$$

Corollary 2.3. *For the formal power series $Z(u)$ in u^{-1} we have*

$$\Delta\colon\ Z(u) \mapsto Z(u)\otimes Z(u), \qquad \varepsilon\colon\ Z(u)\mapsto 1, \qquad \mathrm{S}\colon Z(u)\mapsto Z(u)^{-1}. \tag{2.10}$$

Proof. The square of the antipodal mapping is a coalgebra homomorphism. Hence the images of $T_{ij}(u)$ relative to the two compositions $\Delta\,\mathrm{S}^2$ and $\big(\mathrm{S}^2\otimes\mathrm{S}^2\big)\Delta$ are the same. By Proposition 2.2 these images are respectively equal to

$$\Delta\big(Z(u)^{-1}T_{ij}(u+M-N)\big)$$
$$= \Delta\big(Z(u)^{-1}\big)\sum_{k=1}^{M+N} T_{ik}(u+M-N)\otimes T_{kj}(u+M-N)(-1)^{(\bar{\imath}+\bar{k})(\bar{\jmath}+\bar{k})},$$

and

$$\sum_{k=1}^{M+N} \mathrm{S}^2(T_{ik}(u))\otimes \mathrm{S}^2(T_{kj}(u))(-1)^{(\bar{\imath}+\bar{k})(\bar{\jmath}+\bar{k})}$$
$$= \big(Z(u)^{-1}\otimes Z(u)^{-1}\big)\sum_{k=1}^{M+N} T_{ik}(u+M-N)\otimes T_{kj}(u+M-N)(-1)^{(\bar{\imath}+\bar{k})(\bar{\jmath}+\bar{k})}.$$

By equating the two images of $T_{ij}(u)$ we obtain that

$$\Delta\colon\ Z(u)^{-1}\mapsto Z(u)^{-1}\otimes Z(u)^{-1}.$$

Since the comultiplication Δ is an algebra homomorphism, we get the first statement in (2.10).

The second statement in (2.10) immediately follows from (2.1) and from the definition of the counit homomorphism ε. The third statement follows from the first and the second because the multiplication $\mu\colon \mathrm{Y}(\mathfrak{gl}_{M|N})\otimes\mathrm{Y}(\mathfrak{gl}_{M|N})\to\mathrm{Y}(\mathfrak{gl}_{M|N})$ and the unit mapping $\delta\colon\mathbb{C}\to\mathrm{Y}(\mathfrak{gl}_{M|N})$ satisfy the identity $\mu(\mathrm{S}\otimes\mathrm{id})\Delta = \delta\varepsilon$. Indeed, by applying to the coefficients of the series $Z(u)$ the homomorphisms at the two sides of this identity we get the equality $\mathrm{S}(Z(u))Z(u)=1$. $\square$

In Section 4 we will also use the following observation. For any indices i, j by (1.18) we have

$$\sum_{k=1}^{M+N} T'_{ik}(u)T_{kj}(u)(-1)^{(\bar{\imath}+\bar{k})(\bar{\jmath}+\bar{k})} = \delta_{ij}$$

similarly to (2.9). The collection of last displayed identities can be written as a single relation

$$(Q\otimes 1)T_2^{\sharp}(u)T_1(u) = Q\otimes 1 \tag{2.11}$$

of series in u with coefficients in the algebra $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes 2}\otimes\mathrm{Y}(\mathfrak{gl}_{M|N})$.

Proposition 2.4. *The series $Z(u)$ is invariant under the antiautomorphism* (1.21) *of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$.

Proof. The series $Z(u)$ is equal to the sum at the left-hand side of the relation (2.1) with $i = j$. By applying the antiautomorphism (1.21) to that sum and using Proposition 1.5 we get

$$\sum_{k=1}^{M+N} T'_{ki}(u)T_{ik}(u+M-N)(-1)^{\bar{k}(\bar{\imath}+1)+\bar{\imath}(\bar{k}+1)+(\bar{k}+\bar{\imath})(\bar{\imath}+\bar{k})},$$

which is equal to the sum the left-hand side of the relation (2.2) with $i = j$. □

Corollary 2.5. *The automorphism* (1.24) *of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *maps* $Z(u) \mapsto Z(u)^{-1}$.

Proof. The automorphism (1.24) is the composition of the antiautomorphisms (1.19) and (1.21) of $\mathrm{Y}(\mathfrak{gl}_{M|N})$. Hence the required statement follows from Corollary 2.3 and Proposition 2.4. □

Due to the definitions (1.18) and (2.1) the coefficient of the series $Z(u)$ at u^{-1} is zero. Thus

$$Z(u) = 1 + Z^{(2)}u^{-2} + Z^{(3)}u^{-3} + \cdots \tag{2.12}$$

for certain central elements $Z^{(2)}, Z^{(3)}, \ldots \in \mathrm{Y}(\mathfrak{gl}_{M|N})$. The main result of the present section is

We will use the following proposition.

Proposition 2.6. *For any $r \geqslant 2$ the element $Z^{(r)} \in \mathrm{Y}(\mathfrak{gl}_{M|N})$ has degree $r-2$ relative to the second filtration on $\mathrm{Y}(\mathfrak{gl}_{M|N})$. Its image in the graded algebra $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ is equal to*

$$(1-r)\sum_{i=1}^{M+N} Y_{ii}^{(r-1)}(-1)^{\bar{\imath}}. \tag{2.13}$$

Proof. Let us expand the relation (2.5) of series with coefficients in $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes 2}\otimes \mathrm{Y}(\mathfrak{gl}_{M|N})$ by using the basis of $\mathrm{End}\,\big(\big(\mathbb{C}^{M|N}\big)^{\otimes 2}\big)$ constituted by the elements $E_{ij}\otimes E_{kl}$. By taking $(-1)^{\bar{\imath}+\bar{\jmath}}$ times the terms in the expansion corresponding to the basis element $E_{ij}\otimes E_{ij}$ with any given indices i and j we obtain the relation of series with coefficients in the algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$

$$\begin{aligned} &T_{ij}(u)T'_{ji}(v) + \sum_{k=1}^{M+N} T_{kj}(u)T'_{jk}(v)(u-v-M+N)^{-1}(-1)^{\bar{\imath}} \\ &= T'_{ji}(v)T_{ij}(u)(-1)^{\bar{\imath}+\bar{\jmath}} + \sum_{k=1}^{M+N} T'_{ki}(v)T_{ik}(u)(u-v-M+N)^{-1}(-1)^{\bar{\imath}}. \end{aligned} \tag{2.14}$$

For any i and j denote by $\dot{T}_{ij}(u)$ the formal derivative of the series $T_{ij}(u)$ so that

$$\dot{T}_{ij}(u) = -T_{ij}^{(1)}u^{-2} - 2T_{ij}^{(2)}u^{-3} - \cdots. \tag{2.15}$$

By tending in the relation (2.14) the parameter u to $v+M-N$ we then get

$$\begin{aligned} &T_{ij}(v+M-N)T'_{ji}(v) + \sum_{k=1}^{M+N} \dot{T}_{kj}(v+M-N)T'_{jk}(v)(-1)^{\bar{\imath}} \\ &= T'_{ji}(v)T_{ij}(v+M-N)(-1)^{\bar{\imath}+\bar{\jmath}} + \sum_{k=1}^{M+N} T'_{ki}(v)\dot{T}_{ik}(v+M-N)(-1)^{\bar{\imath}}. \end{aligned} \tag{2.16}$$

Let us now observe that

$$T_{ij}(v+M-N) = T_{ij}(v) + (M-N)\dot{T}_{ij}(v) + O_{ij}(v),$$

where $O_{ij}(v)$ is a certain formal power series in v^{-1} with coefficients in $\mathrm{Y}(\mathfrak{gl}_{M|N})$ such that the coefficient at v^{-r} with $r \geqslant 3$ has degree $r-3$ relative to the second fitration. The coefficient of this series at v^{-r} with any $r \leqslant 2$ is zero. By taking the sum of the relations (2.16) over the indices $i = 1, \dots, M+N$ and then using the definition (2.1) along with this observation we get

$$Z(v) + (M-N)\sum_{k=1}^{M+N} \dot{T}_{kj}(v+M-N)T'_{jk}(v) = \sum_{i,k=1}^{M+N} T'_{ki}(v)\dot{T}_{ik}(v+M-N)(-1)^{\bar{\imath}}$$
$$+ \sum_{i=1}^{M+N} T'_{ji}(v)(T_{ij}(v) + (M-N)\dot{T}_{ij}(v) + O_{ij}(v))(-1)^{\bar{\imath}+\bar{\jmath}}.$$

By using the definition (1.18) the latter relation can be rewritten as

$$\begin{aligned} Z(v) + (M-N)\sum_{k=1}^{M+N} \dot{T}_{kj}(v+M-N)T'_{jk}(v) \\ = \sum_{i,k=1}^{M+N} T'_{ki}(v)\dot{T}_{ik}(v+M-N)(-1)^{\bar{\imath}} \\ + 1 + \sum_{i=1}^{M+N} T'_{ji}(v)((M-N)\dot{T}_{ij}(v) + O_{ij}(v))(-1)^{\bar{\imath}+\bar{\jmath}}. \end{aligned} \tag{2.17}$$

For any indices i and j the leading term of the series $T'_{ij}(v)$ is δ_{ij} while for every $r \geqslant 1$ the coefficient of this series at v^{-r} has degree $r-1$ relative to the second filtration on $\mathrm{Y}(\mathfrak{gl}_{M|N})$. Furthermore for any given $r \geqslant 1$ the coefficients at v^{-r} of the series $\dot{T}_{ij}(v)$ and $\dot{T}_{ij}(v+M-N)$ have the same image in the graded algebra $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$, see (2.15). Therefore by taking the coefficients at v^{-r} with any $r \geqslant 2$ in the relation (2.17) the image of $Z^{(r)}$ in $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ equals

$$-(M-N)\sum_{k=1}^{M+N}(1-r)Y_{kj}^{(r-1)}\delta_{jk} + \sum_{i,k=1}^{M+N}\delta_{ki}(1-r)Y_{ik}^{(r-1)}(-1)^{\bar{\imath}}$$
$$+ \sum_{i=1}^{M+N}\delta_{ji}(M-N)(1-r)Y_{ij}^{(r-1)}(-1)^{\bar{\imath}+\bar{\jmath}} = (1-r)\sum_{i=1}^{M+N}Y_{ii}^{(r-1)}(-1)^{\bar{\imath}}.$$

□

In our proof of Theorem 0.1 we will also use the following general proposition. Let $\mathfrak{a}$ be any finite-dimensional Lie superalgebra over $\mathbb{C}$. Take the polynomial current Lie superalgebra $\mathfrak{a}[u]$.

Proposition 2.7. *Suppose the centre of the Lie superalgebra $\mathfrak{a}$ is trivial. Then the centre of the universal enveloping algebra $\mathrm{U}(\mathfrak{a}[u])$ is also trivial, that is equal to $\mathbb{C}$.*

Proof. Consider adjoint action of the Lie superalgebra $\mathfrak{a}[u]$ on its supersymmetric algebra. By Poincaré–Birkhoff–Witt theorem for $\mathrm{U}(\mathfrak{a}[u])$ it suffices to show that the space of invariants of this action is trivial, see [5, Theorem 5.15].

Let A be an element of the supersymmetric algebra of $\mathfrak{a}[u]$ invariant under the adjoint action. Let $K = \dim \mathfrak{a}$. Choose any $\mathbb{Z}_2$-homogeneous basis $f_1, \dots, f_K$ of $\mathfrak{a}$ and write the Lie brackets as

$$[f_p, f_q] = \sum_{r=1}^{K} C_{pq}^r f_r,$$

where $C_{pq}^r \in \mathbb{C}$ are the structure constants. Let d be the maximal degree such that A depends on at least one of the basis elements $f_1u^d, \dots, f_Ku^d$ of $\mathfrak{a}[u]$. Write

$$A = \sum_{m_1,\dots,m_K} A_{m_1\dots m_K} \big(f_1u^d\big)^{m_1} \cdots \big(f_Ku^d\big)^{m_K}, \tag{2.18}$$

where the coefficients $A_{m_1\dots m_K}$ are certain elements of the supersymmetric algebra which do not depend on $f_1u^d, \dots, f_Ku^d$. For each $p = 1, \dots, K$ let $\bar{p}$ be the $\mathbb{Z}_2$-degree of the basis element f_p of $\mathfrak{a}$. We allow $m_1, \dots, m_K$ in (2.18) to range over $0, 1, 2, \dots$ but assume that $A_{m_1\dots m_K} = 0$ if $m_p > 1$ for at least one index p with $\bar{p} = 1$.

For each $p = 1, \dots, K$ we have the equation $\mathrm{ad}(f_pu)(A) = 0$ in the supersymmetric algebra of $\mathfrak{a}[u]$. In particular, the component of the left-hand side of this equation that involves any of the basis elements $f_1u^{d+1}, \dots, f_Ku^{d+1}$ must be zero. Thus

$$\sum_{m_1,\dots,m_K} A_{m_1\dots m_K} \sum_{q,r=1}^{K} m_q C_{pq}^r \big(f_1u^d\big)^{m_1} \cdots \big(f_qu^d\big)^{m_q-1} \cdots \big(f_Ku^d\big)^{m_K} f_ru^{d+1}(-1)^{h_q} = 0,$$

where we used the notation

$$h_q = \sum_{s=1}^{q-1} \bar{p}\bar{s}m_s + \sum_{s=q+1}^{K} \bar{r}\bar{s}m_s. \tag{2.19}$$

If follows that for any non-negative integers $m'_1, \dots, m'_K$ we have the equations

$$\sum_{q=1}^{M} A_{m'_1\dots m'_q+1\dots m'_K}(m'_q+1)C_{pq}^r(-1)^{h'_q} = 0, \qquad \text{where} \quad p, r = 1, \dots, K. \tag{2.20}$$

Similarly to (2.19) here we used the notation

$$h'_q = \sum_{s=1}^{q-1} \bar{p}\bar{s}m'_s + \sum_{s=q+1}^{K} \bar{r}\bar{s}m'_s.$$

Let us now fix any non-negative integers $m'_1, \dots, m'_K$ and observe that the elements

$$f'_q = (m'_q+1)f_q(-1)^{h'_q} \qquad \text{with} \quad q = 1, \dots, K$$

also form a basis of $\mathfrak{a}$. Since the centre of the Lie superalgebra $\mathfrak{a}$ is trivial, the system

$$\Big[f_p, \sum_{q=1}^{K} w_q f'_q\Big] = 0 \qquad \text{with} \quad p = 1, \dots, K$$

of linear equations on $w_1, \dots, w_K \in \mathbb{C}$ has only zero solution. This system can be written as

$$\sum_{q=1}^{K} w_q(m'_q+1)C_{pq}^r(-1)^{h'_q} = 0, \qquad \text{where} \quad p, r = 1, \dots, K.$$

Hence by comparing the latter system with (2.20) we obtain that $A_{m'_1\dots m'_q+1\dots m'_K} = 0$ for each index $q = 1, \dots, K$. It follows that $A \in \mathbb{C}$ and that $d = 0$ in particular. □

We can now prove Theorem 0.1. Due to Proposition 2.1 the elements

$$Z^{(2)}, Z^{(3)}, \dots \in \mathrm{Y}(\mathfrak{gl}_{M|N})$$

are central. Therefore it suffices to prove that the images of these elements in the graded algebra $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ are free generators of its centre, see Proposition 2.6. By Theorem 1.8 the graded algebra is isomorphic to the universal enveloping algebra

$\mathrm{U}(\mathfrak{gl}_{M|N}[u])$ via (1.31). Under this isomorphism the element (2.13) of $\mathrm{gr}'\,\mathrm{Y}(\mathfrak{gl}_{M|N})$ with any $r \geqslant 2$ corresponds to the element

$$(r-1)\sum_{i=1}^{M+N} e_{ii}u^{r-2} \in \mathrm{U}(\mathfrak{gl}_{M|N}[u]).$$

Let us demonstrate that all the latter elements are free generators of the centre of $\mathrm{U}(\mathfrak{gl}_{M|N}[u])$. They are algebraically independent by the Poincaré–Birkhoff–Witt theorem for $\mathrm{U}(\mathfrak{gl}_{M|N}[u])$, see [5, Theorem 5.15]. To show that they generate the centre of $\mathrm{U}(\mathfrak{gl}_{M|N}[u])$ consider the quotient of $\mathrm{U}(\mathfrak{gl}_{M|N}[u])$ by the ideal they generate. This quotient is isomorphic to the universal enveloping algebra $\mathrm{U}(\mathfrak{a}[u])$ where the Lie superalgebra $\mathfrak{a}$ is the quotient of $\mathfrak{gl}_{M|N}$ by the span of the element (1.37). The centre of the Lie superalgebra $\mathfrak{a}$ is trivial. Hence the centre of $\mathrm{U}(\mathfrak{a}[u])$ is also trivial, see Proposition 2.7. This argument completes our proof of Theorem 0.1.

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